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The extraction of work from hidden degrees of freedom

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Thermodynamics establishes that information acquired through measurement can be converted into work, as exemplified by Maxwell's demon and Szilard engines. Most experimental realizations of information engines, however, implicitly assume Markovian environments, in which information exchanged with the surroundings is irreversibly lost. Many physical systems instead exhibit environmental memory, with hidden degrees of freedom retaining correlations with the history of the system and giving rise to non-Markovian dynamics. Whether and how such concealed memory can be harnessed as a thermodynamic resource has remained an open question. Here we experimentally demonstrate the extraction of work from environmental memory. Using time-resolved measurements on an optically trapped Brownian particle in equilibrium, we implement a time-delayed double-measurement protocol that retrieves information through backflow from hidden bath degrees of freedom. We show that this information backflow alters the relaxation dynamics, can be quantified independently of initial state effects and—when appropriately exploited—enhances work extraction. Notably, we identify regimes in which the extracted work exceeds the energy stored in the observable degree of freedom alone. Our results establish environmental memory as an experimentally accessible thermodynamic resource and reveal how non-Markovian dynamics can be systematically explored to improve the performance of information engines operating in time-correlated environments.

Maxwell's famous *Gedankenexperiment* illustrates that information about the microscopic fluctuations of an equilibrium system can be converted into work extracted from a heat bath. The hypothetical demon thereby reveals a fundamental connection between information and energy conversion at small scales. This insight is formalized in information thermodynamics, which unifies work, entropy production and information within a consistent non-equilibrium theory[1–7].

A cornerstone of information thermodynamics is a generalization of the second law of thermodynamics, first derived by Sagawa and Ueda, which bounds the average net work W that can be extracted from a system during a measurement-based feedback protocol[8,9]:

$$W \leq -\Delta F + k_{\mathrm{B}} T I(Z : M). \tag{1}$$

Here ΔF is the free energy difference between the initial and final equilibrium states associated with the extraction protocol, k_{B} is the Boltzmann constant and T is the temperature of a weakly coupled heat bath[5,10]. The mutual information $I(Z:M)$ quantifies the average information gained about the microscopic system state Z from a measurement with possible outcomes $M \in \Omega_M$. The state Z represents the true physical state of the system, which fully determines its energetics and dynamics but is only partially accessible in practice. A measurement provides incomplete information about Z, reflecting finite resolution, noise,

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coarse-graining or unobservable degrees of freedom. Equation (1) shows that information acquired through measurement constitutes a thermodynamic resource, enabling further work extraction on average.

Experimental realizations and theoretical studies of Szilard engines, tests of the Landauer bound[11–16] and verifications of fluctuation theorems have provided direct evidence of information-to-work conversion[17–21]. Most studies assume Markovian, that is memoryless, environments, where exchanged information is irreversibly lost. Often, however, this assumption breaks down: disordered materials, (bio) polymer solutions, periodically driven many-body systems, critical and active media but also open quantum mechanical systems typically exhibit pronounced temporal correlations that render the dynamics non-Markovian[22–28]. Microscopically, such non-Markovian effects originate from hidden environmental degrees of freedom (hDoFs) that retain information about past system–environment interactions and thereby influence subsequent dynamics and energetic exchanges. Only recently have theoretical studies started to formalize this perspective in the context of information engines operating with temporal correlations and information backflow[29,30]. Complementary experimental works have explored specific consequences, such as delayed feedback control and energy recuperation in viscoelastic environments[31,32].

In general, the presence of hDoFs pose a central challenge for theoretical frameworks, as they complicate both the interpretation of entropy production and the applicability of standard information-thermodynamic bounds[33–37]. From the perspective of a Maxwell demon or feedback controller being restricted regarding its observables, the system, therefore, seems opaque, that is, only partially accessible. Moreover, environmental memory encodes information about the history of the system and, thus, constitutes a thermodynamic resource. This naturally raises the question of whether and to what extent such concealed memory can be harnessed for work extraction.

Previous theoretical work has suggested that this is, in principle, possible when the operations of the demon are precisely timed with respect to the system–bath dynamics[36]. We experimentally realized such a mechanism using time-resolved position measurements of an optically trapped Brownian particle in a viscoelastic fluid. Rather than relying on microscopic control or detailed knowledge of the environment, our approach uses delayed measurement pairs to access hDoFs operationally and in a model-independent way. This enables information stored in environmental memory to be retrieved and converted into useful work. More broadly, our results establish information backflow as an experimentally accessible resource for enhancing the performance of information engines in non-Markovian environments.

To elucidate the role of memory in work extraction, it is instructive to revisit equation (1). If the state of the system Z is taken to comprise an observable degree of freedom X together with all environmental hDoFs Y, the dynamics becomes Markovian in the extended state space $Z=\{X,Y\}$, and equation (1) formally applies with the mutual information $I((X,Y):M)$. Throughout, we assume isothermal conditions and restrict to protocols with $\Delta F=0$ (which is the case for any cyclic machine), allowing us to focus exclusively on information-driven work extraction.

It can be generally shown (Methods) that $I((X,Y):M)=I(X:M)$, implying that a single measurement cannot access thermodynamically useful information about environmental memory. Consequently, equation (1) remains unchanged compared with the Markovian case, despite the presence of hDoFs.

Further insight is obtained by considering work extraction conditioned on a specific measurement outcome $M=m$ of an observable variable $x\in X$. The measurement transforms the equilibrium distribution $\rho_{\mathrm{eq}}(x)$ to the conditional distribution $\rho_m(x)=\rho(x\,|\,m)$, and the extractable work obeys[5,38,39]:

$$W_m \leq k_{\mathrm{B}}T\,\mathcal{I}_m(Z), \tag{2}$$

with the relative entropy or Kullback–Leibler (KL) divergence $\mathcal{I}_m(Z)=D_{\mathrm{KL}}[\rho_m(z)||\rho_{\mathrm{eq}}(z)]$. Averaging this bound over m, with P_m the probability of measurement outcome m, recovers equation (1), as the mean relative entropy is the mutual information $\sum_{\Omega_M}P_m\mathcal{I}_m(Z)=I(Z:M)$. The outcome-resolved bound in equation (2) immediately shows why hDoFs cannot be exploited through a single-time measurement: owing to conditional independence ($M\perp Y\,|\,X$), one has $D_{\mathrm{KL}}[\rho_m(x,y)\,||\,\rho_{\mathrm{eq}}(x,y)]=D_{\mathrm{KL}}[\rho_m(x)\,||\,\rho_{\mathrm{eq}}(x)]$. This accords with physical intuition: memory manifests through temporal correlations, which are inaccessible to a single snapshot measurement. Exploiting memory, therefore, requires measurements at several times.

Repeated measurements are generally beneficial in the presence of measurement uncertainty (for example, measurement error or binning), as they increase the available information about the system state. Consequently, even in Markovian systems, repeated measurements can enhance the amount of extractable work (equation (1))[3,6]. For protocols consisting of a measurement followed by feedback, however, the total extractable work from an initially equilibrium system is fundamentally bounded by the excess free energy in the post-measurement state relative to equilibrium[1]. In systems with memory, environmental hDoFs can contain further excess free energy and lead to information backflow from the environment to the system, thereby, in principle, allowing work extraction beyond the Markovian limit (Methods; specifically, equations (14) and (15) show how both bounds, equations (1) and (2), are changed, respectively). To experimentally access these memory effects, we focus on a minimal measurement protocol capable of probing non-Markovian dynamics: two consecutive binary measurements (left and right). As we demonstrate, this minimal set-up already suffices to retrieve information from memory and convert it into useful work.

As an experimentally accessible and well-controlled model system exhibiting environmental hDoFs, we studied a colloidal silica particle of diameter 2.7 μm suspended in a viscoelastic micellar solution composed of an equimolar 8-mM mixture of cetylpyridinium chloride monohydrate (99%) and sodium salicylate (99.5%) (Methods). At room temperature, the surfactants formed a dynamical micellar network with a single structural relaxation time τ_{r} (refs. 40,41). For our concentrations, we had $\tau_r\approx 6$ s (Supplementary Note 2). As a result, a colloidal particle suspended in this solution exhibited pronounced non-Markovian behaviour[42,43]. Because the micellar network dynamics was not directly observable in our experiments, the state of the micellar network is considered as a hDoF storing information about past particle–bath interactions. The colloid was confined by optical tweezers in a harmonic potential $V(x)=\frac{1}{2}\kappa(x-\lambda)^2$, with x the particle position, λ the trap centre and κ the trap stiffness. Particle trajectories were recorded at a rate 100 s^{-1} and spatial resolution of ±5 nm (Fig. 1a).

The particle position was sampled under equilibrium conditions, as seen by the stationary distribution shown in Fig. 1b. At each measurement time $t=t_i$, the position was compared with a threshold x_{th}, assigning the state $m=|0\rangle$ for $X<x_{\mathrm{th}}$ and $m=|1\rangle$ otherwise. The temporal decay of each such single-measurement outcome was characterized by the corresponding conditional distribution $\rho_{|i\rangle}(x,t)$ (Fig. 1b). Note that a positional measurement does not alter the microscopic state of a classical system, rather it prepares a different statistical ensemble conditioned by the measurement outcome. Selecting a given outcome (for example, $X>x_{\mathrm{th}}$) truncated the equilibrium Boltzmann distribution and created a genuine non-equilibrium state (Fig. 1b).

To access information about the full system, including hDoFs, we, therefore, performed a second measurement at $t_j=t_i+\delta t$. This defined a conditional state $|ij\rangle$ with possible outcomes $m\in\{|00\rangle,|01\rangle,|10\rangle,|11\rangle\}$, characterized by conditional distributions $\rho_{|ij\rangle}(x,t)$ (Fig. 1b). This double-measurement selectively correlated the observed particle configuration with specific thermal excitations of hDoFs, which subsequently relax on their intrinsic timescale τ_{r}. The coupled particle–network relaxation was indirectly reflected in the time evolution of the average potential energy of the particle:

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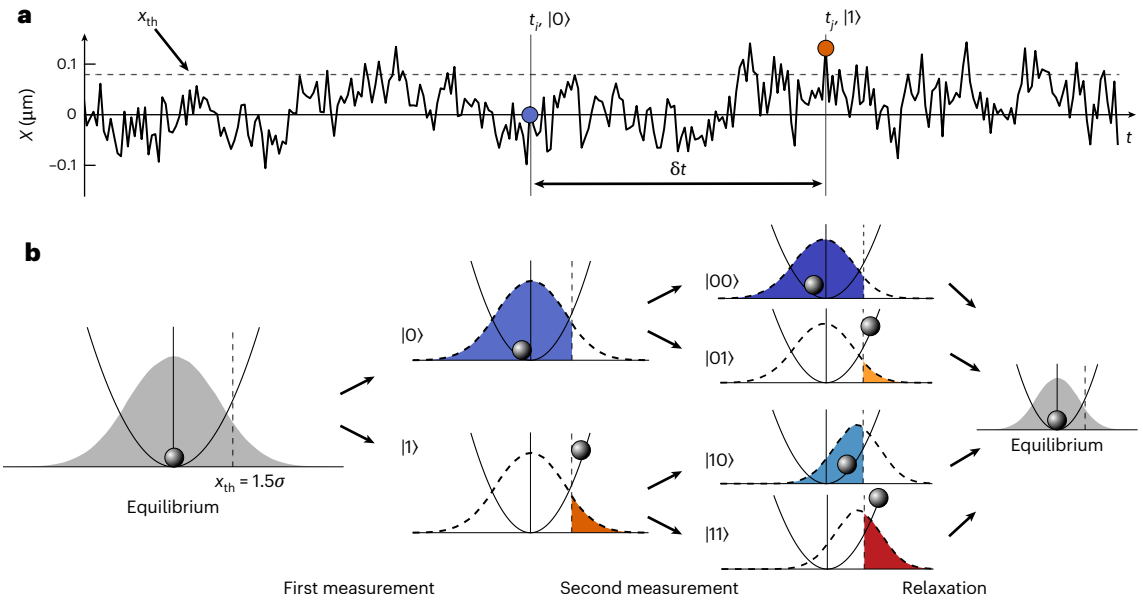


Fig. 1 | Two-measurement protocol and state-conditioned particle distributions. a, Measurement protocol illustrated using an experimental (subsampled for clarity) particle trajectory $x(t)$ (black) corresponding to the state $|01\rangle$. At $t = t_i$ and $t_j = t_i + \delta t$, the particle position is compared with a fixed threshold x_{th}. Positions below and above the threshold are indicated by the blue and orange symbols and define the states $|0\rangle$ and $|1\rangle$, respectively. **b**, Left: equilibrium position distribution $\rho_{eq}(x)$ (grey) in the harmonic potential $V(x)$ (solid black). The vertical dashed line marks the threshold x_{th}. Middle: the first measurement truncates the equilibrium distribution (black dashed line) and prepares the conditional states $\rho_{|0\rangle}(x, 0)$ (blue) and $\rho_{|1\rangle}(x, 0)$ (orange). Right: a second measurement at $t_j = t_i + \delta t$ prepares a further refined conditional ensemble. The $\rho_{|0\rangle}(x, \delta t)$ distribution (dashed line) is then split into $\rho_{|00\rangle}(x, 0)$ (blue) and $\rho_{|01\rangle}(x, 0)$ (light orange). The $\rho_{|1\rangle}(x, \delta t)$ (dashed line) distribution splits into $\rho_{|10\rangle}(x, 0)$ (teal) and $\rho_{|11\rangle}(x, 0)$ (red). All distributions subsequently relax back to equilibrium (grey). The solid spheres correspond to the mean particle position of the corresponding probability distributions.

$$\bar{V}_m(t) = \int \mathrm{d}x\, \rho_m(x, t)\, V(x). \tag{3}$$

Figure 2a shows the state-conditioned relaxation curves $\bar{V}_{|ij\rangle}(t)$ for $\delta t = 1$ s, each obtained from 300 s of trajectory data and averaged over 10^3 realizations. The strong dependence on the initial state $|i\rangle$ for a fixed final state $|j\rangle$ demonstrates that double measurements are sensitive to environmental memory. Most notably, the non-monotonic behaviour of the $|10\rangle$ state provides direct experimental evidence of energy transfer from hDoFs to the particle. By contrast, for large delays $\delta t = 200$ s, the relaxation curves no longer depend on the initial state $|i\rangle$ but only on the final state $|j\rangle$. This indicates that the hDoFs have equilibrated and no longer retain memory of the earlier state (Fig. 2b). Consequently, the relaxation became monotonic and approached the average equilibrium potential energy ($0.5 k_B T$), from both below and above. Memory effects are also expected to vanish in the limit $\delta t \to 0$. This is confirmed by the inset of Fig. 2b, which shows the relaxation behaviour $\bar{V}_{|j\rangle}(t)$ after a single measurement (corresponding to $\delta t = 0$ s). As expected, these curves are nearly identical to the state-conditioned relaxations with identical final state $|j\rangle$ for $\delta t = 200$ s.

To provide a simple physical interpretation, we also considered a minimal micromechanical model (Supplementary Note 4), in which the hDoF of the non-Markovian environment is represented by a fictitious bath particle elastically coupled to the experimentally observed colloidal particle[32,43,44]. The resulting relaxation curves are in good agreement with the experimental data and, in particular, also reproduce the characteristic energy backflow of the $|10\rangle$ state (Supplementary Fig. 3a).

To experimentally quantify the extra information gained from a time-delayed second measurement, we compared the conditional distributions $\rho_{|ij\rangle}(x, t)$ and $\rho_{|j\rangle}(x, t)$ for $t \geq 0$. Both ensembles ended up in the same final state $|j\rangle$, but $\rho_{|ij\rangle}(x, t)$ additionally retained information about the earlier outcome $|i\rangle$. Because these distributions already differed at $t = 0$, any direct comparison would mix effects due to different initial conditions with genuine memory contributions. We, therefore, constructed a new, history-blind, control distribution $\tilde{\rho}_{|ij\rangle}(x, t)$ by randomly subsampling trajectories from the single-measurement ensemble $\rho_{|j\rangle}(x, t)$ at $t = 0$, such that the empirical distribution of particle positions at $t = 0$ matched that of the corresponding two-measurement state (Supplementary Note 3). By construction, this ensured that

$$\tilde{\rho}_{|ij\rangle}(x, 0) = \rho_{|ij\rangle}(x, 0). \tag{4}$$

For a memoryless system, $\tilde{\rho}_{|ij\rangle}(x, t) = \rho_{|ij\rangle}(x, t)$, for all x and all $t \geq 0$. Starting from such control ensembles, we then compared the averaged temporal evolution of observables obtained from single- and two-measurement outcomes, thereby isolating the contribution arising from information stored in hDoFs, which we refer to as information backflow. Any difference between the two ensembles also directly reflects information gained from the second, time-delayed measurement.

We characterized the time-dependent information content using the KL divergence with respect to the equilibrium distribution, $\mathcal{I}_{|ij\rangle}(t) = D_{KL}[\rho_{|ij\rangle}(x, t) || \rho_{eq}(x)]$. Applying the same construction to the control distribution yielded the corresponding control information $\tilde{\mathcal{I}}_{|ij\rangle}(t)$. This quantity would bound the extractable work, if the system was Markovian, or, equivalently, if the system was non-Markovian but only a single measurement was done (equation (2)). By construction, the two ensembles have identical initial information, $\tilde{\mathcal{I}}_{|ij\rangle}(0) = \mathcal{I}_{|ij\rangle}(0)$.

In a Markovian system, the instantaneous state fully determines the future evolution, implying identical information decay for both ensembles, so that $\tilde{\mathcal{I}}_{|ij\rangle}(t) = \mathcal{I}_{|ij\rangle}(t)$ for all t (Supplementary Fig. 6a). The difference between $\tilde{\mathcal{I}}_{|ij\rangle}(t)$ and $\mathcal{I}_{|ij\rangle}(t)$ observed at $\delta t \approx 1$ s (Fig. 3a,b), therefore, provides direct evidence for a backflow of information about the observable x stored in hDoFs of the micellar network, which is revealed only through the evolution of the system after the second, time-delayed measurement.

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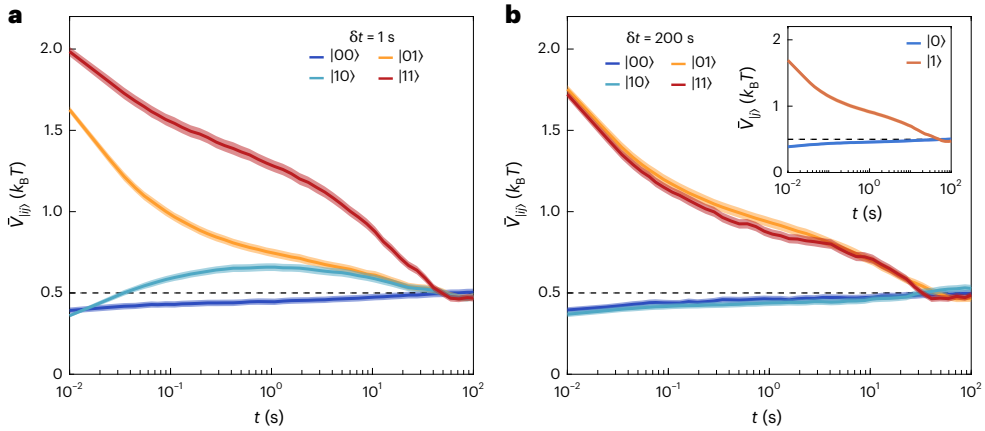


Fig. 2 | Memory-dependent relaxation: state-conditioned relaxation of the averaged particle potential energy. a, State-conditioned relaxation curves $\bar{V}_{|ij\rangle}(t)$ following a two-measurement protocol at $\delta t = 1$ s, showing pronounced differences between states with identical final outcome j but different initial states i. **b**, For $\delta t = 200\,\mathrm{s} \gg \tau_r$, the relaxation curves collapse for fixed j, demonstrating that the dynamics becomes independent of the initial state $|i\rangle$ due to the equilibration of the hDoFs. Inset: relaxation following a single measurement ($\delta t = 0$ s), closely matching the large-δt limit, as memory effects vanish in both cases. All curves relax towards the equilibrium value $0.5\,k_BT$. The shaded bands correspond to the standard error of the mean (s.e.m.) calculated using a bootstrapping procedure (Methods).

To obtain the average time-resolved information at time t after a double measurement, we weighted each state $|ij\rangle$ by its probability P_{ij}:

$$I(t) = \sum_{ij} \mathrm{P}_{ij} \mathcal{I}_{|ij\rangle}(t). \tag{5}$$

As shown in Fig. 3c, for $\delta t = 1$ s, $I(t)$ decayed more slowly than the corresponding measure $\tilde{I}(t)$ despite identical initial values, demonstrating a history-dependent contribution beyond the instantaneous state. Inspired by ref. 45, we captured this excess contribution using the integrated order parameter:

$$\Phi(\delta t) = \sum_{ij} \mathrm{P}_{ij} \int_0^\infty \left[\mathcal{I}_{|ij\rangle}(t) - \tilde{\mathcal{I}}_{|ij\rangle}(t)\right] dt, \tag{6}$$

which quantifies the net information stored in hDoFs that influences the subsequent dynamics of the system. A value $\Phi(\delta t) \approx 0$ indicates memoryless (Markovian) behaviour, for which the relaxation is fully determined by the instantaneous particle state (Supplementary Fig. 6b). By contrast, $\Phi(\delta t) \neq 0$ indicates the influence of hDoFs, whose internal states affect the relaxation speed towards equilibrium. This interpretation also holds at the level of individual measurement-conditioned states and is reflected in the sign of $\mathcal{I}_{|ij\rangle}(t) - \tilde{\mathcal{I}}_{|ij\rangle}(t)$ (Fig. 3a–c). By varying the measurement interval δt, we thus mapped the temporal window over which memory affects the particle. Importantly, a non-zero contribution to $\Phi(\delta t)$ corresponds to a delayed decay of information and coincides with regimes where energy is transiently transferred from hDoFs back to the particle, establishing a direct link between information backflow and the energy backflow from the non-Markovian environment. Figure 3d shows the state-resolved contributions $\mathrm{P}_{ij}\Phi_{|ij\rangle}(\delta t) = \mathrm{P}_{ij} \int_0^t \left[\mathcal{I}_{|ij\rangle}(t) - \tilde{\mathcal{I}}_{|ij\rangle}(t)\right] dt$ Despite pronounced differences in their shapes, the extreme values of all curves (except $\Phi_{|00\rangle}$) occurred almost at the same delay time $\delta t \approx 1$ s, which was set by the timescales of the decay of hDoFs and the particle relaxation in the optical trap. Numerical simulations of the non-Markovian model reproduced this behaviour, yielding an order parameter $\Phi(\delta t)$ in good agreement with the experimental results (Supplementary Fig. 3b). As shown below, a similar state-resolved structure of $\Phi_{|ij\rangle}(\delta t)$ is reflected in the corresponding contributions to the extracted work, highlighting Φ as a predictive indicator for the efficiency of extracting work from hDoFs.

To test experimentally whether information retrieved from hDoFs can be converted into work, we compared with the fundamental information-theoretic bound in equation (2). We begin by rewriting this relation in terms of the decrease in the non-equilibrium free energy[5]:

$$W_m \leq k_B T \mathcal{I}_m(Z) = \mathcal{F}_{\mathrm{neq}}[\rho_m(z), V] - F_{\mathrm{eq}}, \tag{7}$$

where F_{eq} is the free energy of the final equilibrium configuration and $\mathcal{F}_{\mathrm{neq}}[\varrho; U] = \langle U\rangle_\varrho - \sum_x k_B T \varrho(x) \ln \varrho(x)$ denotes the non-equilibrium free energy for a state ϱ under external potential U (ref. 1).

We now consider the experimental protocol used throughout this work, consisting solely of rigid translational shifts of the external potential by an amount Λ over a duration t_f: $V^\Lambda = V(x - \Lambda)$. Under these conditions, equation (2) simplifies considerably: for Markovian systems, the entropic contribution to the free energy difference in equation (7) vanishes, allowing the information-theoretic bound to be replaced by the tighter energetic bound:

$$W_m \leq \underbrace{\langle V(0)\rangle_m - \min_\Lambda[\langle V^\Lambda(0)\rangle_m]}_{=: W_m^*} \leq \bar{V}_m(0), \tag{8}$$

where $\bar{V}_m(0) = \langle V(0)\rangle_m$ is the mean potential energy of the post-measurement state of x (equation (3)) before relaxation takes place. For convenience, we denote the corresponding energetic bound by W_m^*. By contrast, in systems with memory, the extractable work is not bounded by equation (8), as hDoFs can store further excess free energy generated by thermal excitations, which becomes accessible during feedback.

In our experiment, we considered a particle confined in a single-well harmonic potential with stiffness κ, which serves as a minimal and well-controlled reference setting. Here equation (8) simplifies to (Methods):

$$W_m^* = V(\langle X\rangle_m) = \frac{\kappa}{2}\langle X\rangle_m^2. \tag{9}$$

Thus, the extractable work in the Markovian reference system is bounded by the potential energy of the average, post-measurement state.

We conducted the following protocol. After the second measurement, at time $t = 0$, the optical trap was translated at constant velocity $\dot{\lambda}$ over a duration $t_f = 3$ s (Supplementary Note 6). The fluctuating work performed during the extraction process is given by

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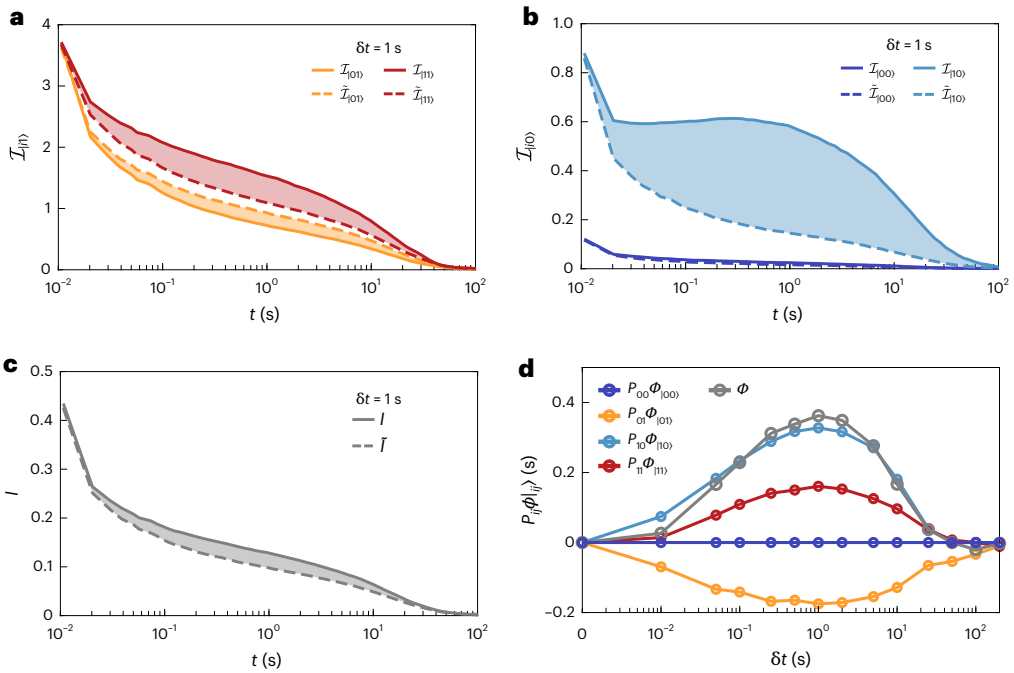


Fig. 3 | Information backflow: information decay and non-Markovian order parameter. a, Time evolution of the KL divergence $\mathcal{I}_{|ij\rangle}(t)$ (solid lines) and $\tilde{\mathcal{I}}_{|ij\rangle}(t)$ (dashed lines) for $\delta t = 1$ s, conditioned on the second measurement outcome $j = 1$. The shaded regions denote enhanced persistence (accelerated decay) of information relative to the adjusted state. Notably, for the state $|01\rangle$, the difference $\mathcal{I}_{|01\rangle}(t) - \tilde{\mathcal{I}}_{|01\rangle}(t)$ is negative, corresponding to an accelerated decay of information. **b**, Same as **a**, but for states conditioned on $j = 0$. **c**, State-averaged information $I(t)$ and $\widetilde{I(t)}$, obtained by weighting individual contributions with their empirical probabilities P_{ij}. The shaded region highlights the net excess information retained at $\delta t = 1$ s. **d**, Integrated order parameters weighted by their relative probabilities $P_{ij}\Phi_{|ij\rangle}(\delta t)$ for individual measurement-conditioned states together with the state-averaged quantity $\Phi(\delta t)$, which show a pronounced maximum at $\delta t \approx 1$ s.

$$w = -\int_0^{t_f} \frac{\partial V(x)}{\partial \lambda}\,\dot{\lambda}\,\mathrm{d}t, \tag{10}$$

where positive values correspond to work extracted from the system. Figure 4b shows the average extracted work after measurement outcomes m, which we denote $W_m(t_f)$. We normalize $W_m(t_f)$ by the energetic bound W_m^* for the Markovian reference system from equation (9). The maximum conversion efficiency $\eta^*_{|11\rangle}(\delta t)$, indicated by markers in Fig. 4a, exhibits a pronounced dependence on the measurement interval δt. The maximum is obtained for $\delta t \approx 1$ s (see inset). Notably, this non-monotonic behaviour closely resembles the trend of the order parameter Φ from Fig. 3d, which shows a maximum around a similar value of δt, indicating maximal information backflow for $\delta t \approx 1$ s. This demonstrates that information retrieved from the second, time-delayed measurement enhances work extraction, a feature that is absent from Markovian environments (Supplementary Fig. 4a).

In addition to correctly identifying an optimal timescale δt between measurements, the non-Markovian order parameter Φ also predicts which states are most favourable for extracting energy from the hDoFs. Among all configurations, $|10\rangle$ yields the largest Φ, reflecting maximal information backflow and the associated recovery of non-equilibrium free energy after the double measurement (Fig. 3d). Consistently, this state produces the highest work output from memory, as captured by the normalized work W_m/W_m^* (Fig. 4b). The extracted work reaches values up to three times $W^*_{|10\rangle}$, substantially exceeding what would be achievable from an equivalent measurement in a Markovian environment. This behaviour indicates that energy extraction in the $|10\rangle$ state is dominated by the hDoFs, with only a minor contribution from the observable potential energy. The condition $W_m/W_m^* > 1$, therefore, provides direct evidence that free energy stored in hDoFs can be converted into work using information obtained solely from the tracer particle. Overall, the ordering of work extraction is captured well by Φ.

When considering the absolute amount of extracted work W_m (Fig. 4b, inset), which combines contributions from the memory as well as the bare potential energy of x of the initial state, a different ordering of the states emerges. This can be partially attributed to the largely different values of $\langle V(0)\rangle_{|ij\rangle}$. State $|11\rangle$ allows the highest absolute value of work to be extracted, owing to a substantial contribution from the observable potential energy $\langle V(0)\rangle_{|11\rangle}$. Although $|01\rangle$ seems similar at first glance, as $\langle V(0)\rangle_{|01\rangle} \approx \langle V(0)\rangle_{|11\rangle}$, the work extracted is considerably smaller. Moreover, $W_{|01\rangle}$ reaches its maximum earlier, consistent with $\Phi_{|01\rangle} < 0$, which indicates that the hDoFs accelerate the relaxation dynamics in this case. Finally, no work can be extracted from the $|00\rangle$ state, for which $W_{|00\rangle}$ remains systematically negative. Numerical simulations reproduced this behaviour and are in good agreement with the experimental results (Supplementary Fig. 4b).

In this work, we have demonstrated work extraction beyond the Markovian limit, focusing on the minimum measurement scheme to access memory—two binary measurements—and an extraction protocol restricted to rigid potential shifts. By using more flexible and optimized extraction protocols or by increasing the number of measurements or their spatial resolution to increase the accessible information, the extractable work could further be enhanced; the fundamental limit set for work extraction is set by the total excess free energy of the (thermally excited) system state at the onset of control (equation (7)).

Our results highlight a general route for identifying and exploiting hDoFs as a thermodynamic resource in systems with intrinsic memory. In this context, time-delayed or repeated measurements provide an operational handle that can be used to infer statistical information about the state of hDoFs from partial observations, without requiring any explicit or direct access to their microscopic dynamics.

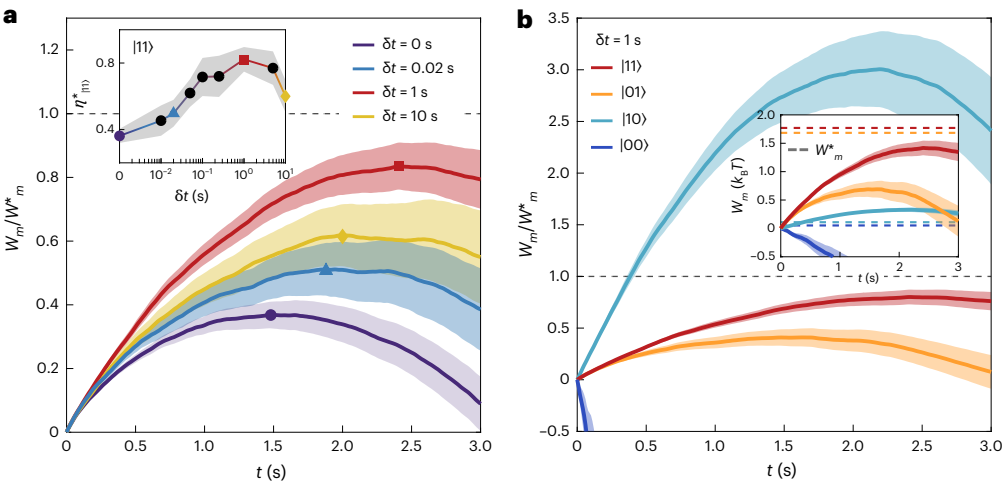


Fig. 4 | Work extraction from hDoFs through two-measurement protocols. **a**, Averaged normalized work $W_m(t)/W^*_m$ along finite-time extraction protocols, extracted from a $|11\rangle$ state and for increasing δt (purple to yellow). The work is normalized by the theoretical optimum for the corresponding Markovian system, W^*_m (equation (9)). Markers denote the maximal normalized work for each interval, $\eta^*_{|11\rangle}$, which peaks at $\delta t = 1$ s (see inset) but remains below unity. **b**, Averaged normalized work W_m/W^*_m extracted for a fixed value $\delta t = 1$ s and states $|11\rangle$ (red), $|01\rangle$ (light orange), $|10\rangle$ (teal) and $|00\rangle$ (blue). For $|10\rangle$, the maximum extracted work clearly exceeds W^*_m, demonstrating the extraction of energy stored within hDoFs. Although the ratio of work that can be extracted from the hDoFs is highest in state $|10\rangle$, the absolute amount of extractable work is highest in $|11\rangle$ (inset). The averages have been obtained from ~100 extraction protocols, and the shaded areas correspond to the s.e.m. obtained from bootstrapping (Methods).

This enables memory-aware control strategies, such as adaptive feedback protocols that tailor control actions to the inferred bath state and allow accelerated relaxation towards equilibrium in time-correlated environments where conventional Markovian shortcuts fail[46].

Beyond control, double measurements also suggest an operational criterion for memory erasure by signalling when all information-bearing hidden modes have relaxed and the dynamics becomes history independent. Beyond classical systems, the storage and backflow of information in unobserved environmental modes are a central theme in the theory of open quantum systems, where non-Markovianity has been widely discussed[27,28]. Our findings thus point towards a unifying perspective in which hDoFs—whether classical or quantum—can be systematically characterized and exploited for information-based control, equilibration and energy conversion in time-correlated environments.

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Methods

Single measurement cannot harvest memory

Here we discuss an abstract set-up in which a work extraction stage starts immediately after one or more measurements completed at time $t = 0$, with no further measurements performed during the extraction stage.

Let the extended state space comprise the observable degrees of freedom X together with all hDoFs Y, which are not directly measurable. All degrees of freedom are assumed to be at the same temperature so that the combined system is initially in equilibrium.

In the extended state space $Z = \{X, Y\}$ the dynamics becomes Markovian, and the generalized second law of information thermodynamics given in equation (1) formally applies with the mutual information $I((X,Y):M) = \iint \rho(x,y,m) \ln \frac{\rho(x,y,m)}{\rho(x,y)\rho(m)}\, dx\, dm$. We specify cases where the equilibrium free energy difference vanishes, $\Delta F = 0$, which is appropriate for rigid potential shifts and any cyclic protocol. We, therefore, omit the free energy term in equation (1) throughout, which, thus, takes the form $W \leq k_B T I((X,Y):M)$.

The measurement apparatus is assumed not to be directly influenced by Y. In an information-theoretical language, the measurement outcome M is generated via a channel $\rho(m \mid x)$ that depends only on X. Consequently[47],

$$\rho(m|x,y) = \rho(m|x), \quad \rho(y|x,m) = \rho(y|x), \tag{11}$$

implying the conditional independence property $Y \perp M \mid X$: $\rho(y,m \mid x) = \rho(y \mid x)\rho(m \mid x)$ (ref. 47). Note that this holds for continuous as well as discretized (threshold or binary) snapshot measurements. It follows directly that

$$\begin{aligned} &I(Y:M|X) \\ &= \int dx\, \rho(x) \iint dy\, dm\, \rho(y,m|x) \ln \frac{\rho(y,m|x)}{\rho(y|x,m)\rho(m|x)} \\ &= \int dx\, \rho(x) \iint dy\, dm\, \rho(y,m|x) \ln 1 = 0, \end{aligned} \tag{12}$$

where we used that $\rho(y,m \mid x) = \rho(y \mid m,x)\rho(m \mid x)$. Using the decomposition of the mutual information: $I((X,Y):M) = I(X:M) + I(Y:M \mid X)$ together with equation (12), we obtain

$$I((X,Y):M) = I(X:M). \tag{13}$$

This implies that equation (1) reduces to the same form as for a Markovian system with no hDoFs.

Now we turn to work extraction after a specific measurement outcome m. The outcome-resolved KL bound in equation (2) reads $W_m \leq k_B T \mathcal{I}_m(X,Y)$, where $\mathcal{I}_m(X,Y)$ denotes the KL divergence $\mathcal{I}_m(X,Y) = D_{KL}[\rho_m(x,y) || \rho_{eq}(x,y)]$ giving the non-equilibrium free energy distance of the post-measurement state from equilibrium. We recall the notation, $\rho_m(x) \coloneqq \rho(x \mid m)$.

Owing to the conditional independence $M \perp Y \mid X$, the posterior over the full system factors as $\rho(x,y \mid m) = \rho(y \mid x)\, \rho(x \mid m)$. Consequently, for any realized m,

$$\begin{aligned} &D_{KL}\left[\rho_m(x,y) \parallel \rho(x,y)\right] \\ &= \iint dy\, dx\, \rho(y|x)\, \rho(x|m) \ln \tfrac{\rho(y|x)\,\rho(x|m)}{\rho(y|x)\,\rho(x)} \\ &= \int dx\, \rho(x|m) \ln \tfrac{\rho(x|m)}{\rho(x)} = D_{KL}\left[\rho_m(x) \parallel \rho(x)\right]. \end{aligned}$$

Thus, $\mathcal{I}_m(X,Y) = \mathcal{I}_m(X)$ and the hDoFs disappear from the outcome-resolved bound in equation (2), too.

Several measurements allow memory to be harvested

Owing to measurement uncertainties from sensor noise or binning, several measurements generally increase the acquired information compared with a single measurement, even for Markovian systems[3,6]. Restricting, for simplicity, to the case with two measurements, the generalized second law of equation (1) then formally involves the informational term $I((X_2,Y_2):(M_1,M_2))$. Here (M_i, X_i, Y_i) denote the random variables at the time of the first measurement $t = -\delta t$ for $i = 1$ and at the second measurement $t = 0$ for $i = 2$, after which the extraction stage is initiated. Although M_i depends directly on only X_i, $i = 1, 2$, the hidden variable Y_2 contains information about the earlier state X_1 through memory. Thus, Y_2 and M_1 are generally not independent conditional on X_2. Therefore, the conditional independence property breaks down for repeated measurements.

Decomposing the mutual information, $I(X_2, Y_2 : M_1, M_2) = I(X_2 : M_1, M_2) + I(Y_2 : M_1, M_2 \mid X_2)$, the generalized second law can now be written as

$$\frac{W}{k_B T} \leq I(X_2 : (M_1, M_2)) + I(Y_2 : (M_1, M_2)|X_2), \tag{14}$$

consistent with ref. 6. This decomposition makes explicit that memory can enhance the extractable work by the extra gain $I(Y_2 : (M_1, M_2) \mid X_2) \geq 0$, which quantifies the information about the hDoFs implicitly revealed by the repeated measurements.

Analogously, for fixed measurement outcomes m_1 and m_2, the outcome-resolved bound (equation (2)) on the extractable work is generally increased compared with Markovian systems. This follows from the monotonicity of relative entropy under marginalization[47], which implies that

$$D_{KL}[\rho_{m1,m2}(x,y) \parallel \rho_{eq}(x,y)] \geq D_{KL}[\rho_{m1,m2}(x) \parallel \rho_{eq}(x)]. \tag{15}$$

We conclude that repeated measurements can, in principle, increase the extractable work beyond the Markovian bound by revealing further information stored in the memory. This enhancement occurs provided that the measurements effectively probe correlations residing in the hDoFs[28].

Work extracted from rigid potential shifts

By equation (2), for a fixed measurement outcome m (of a single or double measurement), the maximum extractable work in an isothermal process is bounded by the KL divergence. For Markovian systems, the latter can equivalently be expressed as the decrease in non-equilibrium free energy of x to the final equilibrium state[5]:

$$\begin{aligned} W_m &\leq k_B T \mathcal{I}_m(X) = k_B T D_{KL}\left[\rho_m(x) \parallel \rho_{eq}(x)\right] \\ &= \mathcal{F}_{neq}[\rho_m(x); V] - F_{eq}, \end{aligned} \tag{16}$$

where $\rho_{eq}(x) \propto e^{-V(x)/(k_B T)}$. The non-equilibrium free energy is given by $\mathcal{F}_{neq}[\varrho; U] = \langle U \rangle_\varrho - \sum_x k_B T \varrho(x) \ln \varrho(x)$ for a state ϱ under external potential U (ref. 1). Note that equation (16) corresponds to equation (7) for the Markovian processes considered here. Equation (16) expresses that no more work can be extracted than the free energy created by conditioning on the measurement outcome. Equality can be achieved for reversible (quasistatic) protocols.

Now we address the specific case of work extraction restricted to rigid translations of the externally controllable potential. For such operations, the general informational bound of equation (16) can be replaced by a tighter, energetic bound. Specifically, we restrict the extraction process to rigid translations:

$$V(x) \to V^\Lambda(x) = V(x - \Lambda), \tag{17}$$

without any change to the shape of V. We choose $V \geq 0$ without loss of generality. As the partition function

$$\mathcal{Z}(\Lambda) = \int \mathrm{d}x\, \mathrm{e}^{-\beta V(x-\Lambda)} = \int \mathrm{d}u\, \mathrm{e}^{-\beta V(u)} \tag{18}$$

is independent of Λ, the free energy $F_{\mathrm{eq}} = -k_{\mathrm{B}}T ln\mathcal{Z}$ is also invariant. Therefore, all accessible equilibrium states have the same equilibrium free energy, that is, $F_{\mathrm{eq}}[V^{\Lambda}]$ is independent of Λ. This further implies that the optimal strategy for work extraction amounts to maximizing the conversion of THE non-equilibrium free energy of the initial state $\mathcal{F}_{\mathrm{neq}}[\rho_m; V]$ into work, before it is lost by relaxation.

But how much can we maximally extract by pure rigid potential translations in the Markovian setting? To get an insight, we decompose the dynamics into infinitesimal time steps $0 \to \Delta t \to 2\Delta t \to \ldots$ with corresponding infinitesimal translations $V^0(x) \to V^1(x) \to V^2(x) \to \ldots$ where $V^i(x) = V^{\Lambda(t)}(x)$ and $\Lambda(0) = 0$. At every time step, one can perform one instantaneous translation step, during which the instantaneous entropy $S[\rho(t)]$ is trivially constant, as ρ is not directly affected by a shift of V. Such an instantaneous shift extracts work from potential energy $W^i = \langle V^{i+1}\rangle_{\rho(t)} - \langle V^i\rangle_{\rho(t)}$. Outside of the instantaneous shifts, $V(t)$ is constant, leading to pure relaxation in a fixed potential, which extracts no work. As non-equilibrium free energy is generally a non-increasing function under thermal relaxation, this part of the dynamics does not increase the capacity for extractable work within future translation steps as it can only reduce it. The decomposition of the dynamics, therefore, indicates that the best strategy is to omit an intermediate relaxation and simply perform one large shift that extracts all the available potential energy.

We conclude that for Markovian processes, rigid extraction protocols of finite duration do not outperform instantaneous potential shifts. This leads to the bound $W_m \leq W_m^*$, specified in equation (8) and consistent with ref. 48.

The energetic bound of equation (8) is general for Markovian rigid potential translations (and any V) and generally tighter than the informational bound of equation (2).

By contrast, in non-Markovian systems, the internal coupling between the visible degree of freedom and the hDoFs introduces other sources of free energy, so that the simple bound of equation (8) (and the paradigm of instantaneous protocols) no longer applies.

Refined bound for harmonic potentials

To evaluate the maximally extractable work from the energetic bound given in equation (8) for a single-well potential, we optimized W_m^* with respect to Λ. This yields the optimal shift of $\Lambda = \langle X\rangle_m$ and corresponding maximum work extraction of

$$W_m^* = V(\langle X\rangle_m) = \frac{k}{2}\langle X\rangle_m^2 \leq \overline{V}_m(0), \tag{19}$$

as given in equation (9). Note that the last inequality follows from the Cauchy–Schwarz inequality.

To cross-check that equation (9) (and, hence, equation (8)) cannot, indeed, be beaten by any finite-time extraction protocol (including quasistatic extraction), we employed the framework of optimal control. To this end, we computed the optimal protocol for shifting the trap to some predefined target position λ_{f} in a given time t_{f} after a measurement with outcome $\rho_m(x)$. This can be done by repeating the calculation of ref. 49 for appropriate initial conditions (see their equation (43) with $v_0 = 0$ and $x_0 = \langle X\rangle_m$), which yields

$$W^*(\lambda_{\mathrm{f}}, t_{\mathrm{f}}) = \frac{k}{2}\langle X\rangle_m^2 - \frac{(\lambda_{\mathrm{f}} - \langle X\rangle_m)^2}{t_{\mathrm{f}} - 2/k}. \tag{20}$$

Further optimizing this result with respect to λ_{f} yields $\lambda_{\mathrm{f}}^* = \langle X\rangle_m$, so that we indeed recover equation (9), independent of t_{f}. This confirms equations (9) and (8).

Sample preparation and optical trapping

The viscoelastic micellar solution was prepared by dissolving 284.6 mg cetylpyridinium chloride monohydrate and 128.1 g sodium salicylate in 104.4 g deionized water, leading to an 8 mM equimolar solution. The mixture was stirred at 45 °C for 24 h and equilibrated at 25 °C before use. Silica tracer particles (diameter 2.7 µm) were dispersed at low volume fraction in 100-µm-thick glass capillaries and imaged at 100 fps using a CMOS camera (Basler a2A2600-64umBAS). A 532-nm laser (Coherent Verdi-V2) was focused through a ×100 oil-immersion objective (Zeiss, numerical aperture 1.3) to create the optical potential. Using an acousto-optic modulator (AA Opto-Electronic) to control the laser intensity, we realized a trap stiffness κ ranging from 0.50 µN m^{-1} to 40 µN m^{-1}. Precise thermal stability was maintained at 25 °C using an objective heater (Okolab).

Feedback protocols and statistical analysis

Feedback control was implemented by evaluating particle positions in real time to trigger automated trap displacements via a piezoelectric stage (Piezoconcept LT3.300). In the single-well geometry, measurement outcomes were defined relative to a spatial threshold $x_{\mathrm{th}} = 70$ nm. To maximize work extraction, specific control sequences were triggered upon detection of the state $|ij\rangle$. For these events, the trap was translated at a constant velocity of $\dot{\lambda}$ (Supplementary Note 6).

Statistical uncertainties were quantified using bootstrap resampling to estimate the error in the mean. For the state-conditioned relaxation curves (Fig. 2), which were subsampled from long continuous trajectories, we generated 500 bootstrap realizations by randomly selecting 10^3 segments. For the work extraction data in Fig. 4c,d, where each curve corresponds to a specific measurement delay δt, bootstrapping was performed by generating 2,000 resampled datasets. Each dataset was constructed by selecting the maximum number of independent trajectories with replacement from the original ensemble. In both cases, the shaded bands represent the standard deviation of the bootstrap distribution, providing a robust estimate of the s.e.m.

Data availability

The processed data used to generate the figures in this study are available via Zenodo at https://doi.org/10.5281/zenodo.19881972 (ref. 50). Owing to the large size and complexity of the raw experimental data, they are available only from the corresponding author upon reasonable request.

Code availability

The computer code developed for the data analysis presented in this work is available via Zenodo at https://doi.org/10.5281/zenodo.19881972 (ref. 50).

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Author contributions

L.M. and F.G. designed the experiments, which were carried out and analysed by L.M. The theory was developed by S.A.M.L. C.B. directed the project. All authors participated in writing the Article.

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